

Andrea Colombo - Algorithms for Massive Datasets

course project - A.A 2025/2026

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Abstract — This report contains the

Table of Contents

1	Introduction	2
1.1	Development Environment	2
2	Dataset Description	2
2.1	Preprocessing Techniques	2
3	Flajolet–Martin Algorithm	2
4	AMS Algorithm	2
5	Conclusion	2
6	Plagiarism and AI Usage Statement	2
	Bibliography	2

1 Introduction

1.1 Development Environment

This project was developed using *Google Colab* as the main computation environment. The Jupyter Notebook submitted alongside this project may require some modifications before being run in a local environment.

2 Dataset Description

The dataset used for this project is the **New York Times Articles & Comments (2020)** [1] dataset.

2.1 Preprocessing Techniques

3 Flajolet–Martin Algorithm

This algorithm was first introduced in 1985 in this paper [2],

4 AMS Algorithm

5 Conclusion

6 Plagiarism and AI Usage Statement

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study. No generative AI tool has been used to write the code or the report content.

Bibliography

- [1] B. Dornel, “New York Times Articles & Comments (2020).” 2021.
- [2] F. Philippe and M. G. Nigel, “Proba bit istic Cou nting Algorithms for Data Base Applications,” 1985.